Fixed-Index Jacobi Spectral Expansion Through Inverse-Square Order

Kirtus Justus (Theoretical Physicist) • 31 October 2026
Port Coquitlam, British Columbia, Canada

Abstract

For the tridiagonal Jacobi matrix commuting with the half-filled finite sine-kernel concentration operator, we prove an expansion of each fixed-index upper-edge eigenvalue through the inverse-square term, with an index-dependent inverse-cube error. The argument begins with an exact finite-dimensional quadratic form and combines compact spectral convergence to a harmonic oscillator, quantitative weighted Taylor bounds for sampled Hermite functions, and two perturbative corrections to a finite Hermite quasimode. The spectral-gap argument also controls the approximation of the corresponding fixed-index eigenvector. The scope is fixed index. The theorem does not control indices growing with the matrix size, reference-modular operator norms, or Wilson–Dirac relative-entropy asymptotics.

1. Exact problem and result

For an even positive integer N, let J_N be the real symmetric $N \times N$ matrix

$$(J_N)_{j,j+1} = (J_N)_{j+1,j} = \frac{(j+1)(N-j-1)}{N}, \qquad 0 \le j \le N-2,$$

with all diagonal entries zero. The off-diagonal entries are strictly positive, so the eigenvalues are simple. Write them in strictly decreasing order as $\lambda_{N,0} > \cdots > \lambda_{N,N-1}$.

Theorem. For each fixed nonnegative integer k, as $N \to \infty$ through even integers,

$$\boxed{\lambda_{N,k} = \frac{N}{2} - \frac{2k+1}{\sqrt{2}} + \frac{6k^2+6k+3}{8N} + \frac{\sqrt{2}(2k+1)(k^2+k+5)}{64N^2} + O_k(N^{-3}).} \tag{1}$$

The remainder constant and the starting dimension may depend on k. All statements below use a fixed k; no uniformity over moving indices is asserted.

2. Discrete form and identification of the oscillator index

Set $h = N^{-1/2}$, $H_N = (N/2)I - J_N$, $y_j = h(j - (N-1)/2)$, and

$$a_{j+1/2} = \frac{1}{4h^2} - (y_j + h/2)^2 = \frac{(j+1)(N-j-1)}{N}.$$

The exterior coefficients $a_{-1/2}$ and $a_{N-1/2}$ vanish exactly. Summing squares gives the exact formula

$$\boxed{\langle v, H_N v\rangle = \sum_{j=0}^{N-2} a_{j+1/2}|v_{j+1} - v_j|^2 + \sum_{j=0}^{N-1}(2y_j^2 + h^2/2)|v_j|^2.} \tag{2}$$

The limiting oscillator is

$$H = -\tfrac{1}{4}\partial_y^2 + 2y^2, \qquad H\psi_k = \varepsilon_k \psi_k, \qquad \varepsilon_k = (2k+1)/\sqrt{2}, \tag{3}$$

with normalized Hermite functions ψ_k. We justify the spectral numbering instead of deriving it from a quasimode alone.

For $u \in \mathcal{S}(\mathbb{R})$, define $(\mathcal{S}_h u)_j = \sqrt{h}\, u(y_j)$. Riemann-sum convergence gives convergence of scalar products and of (2) on any fixed finite span of Hermite functions. The min–max principle therefore yields

$$\limsup_{N\to\infty} \varepsilon_{N,k} \le \varepsilon_k, \qquad \varepsilon_{N,k} = N/2 - \lambda_{N,k},$$

where the discrete energies are ordered increasingly.

For the converse, take normalized vectors $v^{(N)}$ with uniformly bounded energy. Formula (2) implies $\sum_j y_j^2 |v_j^{(N)}|^2 \le C$, so their mass outside $|y| \le R$ is at most C/R^2, uniformly in N. Embed them into $L^2(\mathbb{R})$ by the piecewise constant functions $v_j/\sqrt{h}$ on the cells centered at y_j. These embeddings preserve the ℓ^2 norm exactly. For the bond midpoint $y_{j+1/2} = y_j + h/2$, the coefficient satisfies the quantitative local bound

$$a_{j+1/2} = \frac{1}{4h^2} - y_{j+1/2}^2 \ge \frac{1}{8h^2}, \qquad |y_{j+1/2}| \le R, \quad h^2R^2 \le \tfrac{1}{8}.$$

Consequently, for discrete vectors of energy at most C, $\sum_{|y_{j+1/2}|\le R} |v_{j+1} - v_j|^2/h^2 \le 8C$. The piecewise linear interpolants therefore have uniformly bounded local H^1 seminorms, and their local L^2 distance from the piecewise constant embeddings is $O_R(h)$. Local Rellich compactness plus the uniform tail control gives global strong L^2 precompactness. By lower semicontinuity and the local limit of the bond coefficients, every limit u satisfies

$$\langle u, Hu\rangle \le \liminf_N \langle v^{(N)}, H_N v^{(N)}\rangle$$

in the quadratic-form sense. Apply this to the first $k+1$ discrete energy eigenvectors simultaneously, including all their fixed linear combinations. Their limits remain orthonormal. Continuum min–max then gives $\liminf_N \varepsilon_{N,k} \ge \varepsilon_k$. Thus

$$\boxed{\varepsilon_{N,k} \to \varepsilon_k \quad \text{for every fixed } k.} \tag{4}$$

In particular, each sufficiently small fixed neighborhood of ε_k contains precisely one discrete eigenvalue for sufficiently large N. This fact will identify the eigenvalue located by the quasimode.

3. Uniform Taylor remainder on fixed Hermite spans

For a smooth function sampled at $y = y_j$, the exact finite action is

$$\begin{aligned}(H_N \mathcal{S}_h u)_j/\sqrt{h} = {} & (2y^2 + h^2/2)u(y) \\ & - \left(\frac{1}{4h^2} - y^2 - \frac{h^2}{4}\right)\big[u(y+h) + u(y-h) - 2u(y)\big] \\ & + yh\big[u(y+h) - u(y-h)\big].\end{aligned} \tag{5}$$

This identity holds even at an endpoint, because the coefficient multiplying the exterior value is identically zero.

Define

$$
\begin{aligned}
P &= \partial_y(y^2\partial_y) + \tfrac{1}{2} - \tfrac{1}{48}\partial_y^4, \\
Q &= \tfrac{1}{12}\partial_y^2(y^2\partial_y^2) + \tfrac{1}{12}\partial_y^2 - \tfrac{1}{1440}\partial_y^6.
\end{aligned}
\tag{6}
$$

Taylor lemma. For each fixed finite Hermite span $\mathcal{H}_M = \mathrm{span}\{\psi_0, \dots, \psi_M\}$, and for each fixed $u \in \mathcal{H}_M$,

$$
\|H_N \mathcal{S}_h u - \mathcal{S}_h(Hu + h^2 Pu + h^4 Qu)\|_{\ell^2} \le C_u h^6. \tag{7}
$$

The estimate also holds uniformly on the unit sphere of any fixed finite Hermite span, after enlarging the constant.

Justification. Taylor's theorem with integral remainders gives

$$
\begin{aligned}
u(y+h) + u(y-h) - 2u(y) &= h^2 u'' + \tfrac{h^4}{12}u^{(4)} + \tfrac{h^6}{360}u^{(6)} + O_u(h^8), \\
u(y+h) - u(y-h) &= 2hu' + \tfrac{h^3}{3}u^{(3)} + O_u(h^5),
\end{aligned}
$$

where the remainders admit bounds by finite sums of Schwartz seminorms, including the y^2 weights in (5). The leading coefficient $1/(4h^2)$ turns the h^8 symmetric-difference remainder into $O(h^6)$; the remaining terms are also $O(h^6)$ or smaller in the sampled ℓ^2 norm. The weighted sampled Schwartz norms stay bounded uniformly in h by Riemann-sum estimates. Equating the h^0, h^2, h^4 terms gives exactly H, P, Q. The endpoint treatment in (5) requires no nonphysical boundary value: the combined coefficients of the exterior samples vanish exactly. The discarded Gaussian tails are smaller than any inverse power of N. The same argument may be taken one order further: on fixed Hermite spans,

$$
H_N \mathcal{S}_h u = \mathcal{S}_h\big[(H + h^2 P + h^4 Q + h^6 R_6)u\big] + O_u(h^8),
$$

where the additional *Taylor remainder operator*, used only to cross-check the residual scale, is

$$
R_6 = \frac{y^2}{360}\partial_y^6 + \frac{y}{60}\partial_y^5 + \frac{1}{48}\partial_y^4 - \frac{1}{80640}\partial_y^8.
$$

Every derivative in this display acts on a fixed Schwartz function and its sampled weighted norms are uniformly bounded. This higher expansion does not change the statement of the spectral theorem. □

4. Two perturbative coefficients

Use the Hermite ladder operators with $\alpha = 2\sqrt{2}$:

$$
y = \frac{a + a^\dagger}{\sqrt{2\alpha}}, \qquad \partial_y = \sqrt{\frac{\alpha}{2}}(a - a^\dagger).
$$

For any $k \ge 0$, negative-index terms being absent, the nonzero matrix elements of $P\psi_k$ are

$$
\begin{aligned}
P_{kk} &= -\frac{6k^2+6k+3}{8} =: p_k,\\
P_{k-2,k} &= \frac{(2k-1)\sqrt{k(k-1)}}{12},\\
P_{k+2,k} &= \frac{(2k+3)\sqrt{(k+1)(k+2)}}{12},\\
P_{k-4,k} &= \frac{5}{24}\sqrt{k(k-1)(k-2)(k-3)},\\
P_{k+4,k} &= \frac{5}{24}\sqrt{(k+1)(k+2)(k+3)(k+4)}.
\end{aligned}
\tag{8}
$$

The second diagonal coefficient is

$$
\langle \psi_k, Q\psi_k\rangle = \frac{\sqrt{2}}{36}(4k^3+6k^2+8k+3). \tag{9}
$$

Introduce the finite correction

$$
w_k = \sum_{\ell\neq k} \frac{P_{\ell k}}{\varepsilon_k-\varepsilon_\ell}\psi_\ell, \quad (H-\varepsilon_k)w_k = -(P-p_k)\psi_k.
$$

The inverse-square energy coefficient equals

$$
\begin{aligned}
q_k &= \langle \psi_k, Q\psi_k\rangle + \sum_{\ell\neq k}\frac{|P_{\ell k}|^2}{\varepsilon_k-\varepsilon_\ell}\\
&= -\frac{\sqrt{2}(2k+1)(k^2+k+5)}{64}.
\end{aligned}
\tag{10}
$$

Only the four possible off-diagonal shifts in (8) enter the sum; its simplification is an identity in k. No infinite perturbation sum is required.

5. Second corrected quasimode and proof of the theorem

The combination

$$
R_k = (P-p_k)w_k + (Q-q_k)\psi_k
$$

has zero component on ψ_k, by (10), and belongs to a finite Hermite span. Consequently the finite vector

$$
z_k = -\sum_{\ell\neq k}\frac{\langle \psi_\ell, R_k\rangle}{\varepsilon_\ell-\varepsilon_k}\psi_\ell
$$

satisfies $(H-\varepsilon_k)z_k = -R_k$. Let

$$
U_{N,k} = \mathcal{S}_h(\psi_k + h^2 w_k + h^4 z_k), \qquad \mathcal{E}_{N,k} = \varepsilon_k + h^2 p_k + h^4 q_k.
$$

Insert these expressions into (7). The coefficients of h^0, h^2, h^4 cancel separately, leaving

$$
\|(H_N - \mathcal{E}_{N,k})U_{N,k}\|_{\ell^2} \le C_k h^6, \qquad \|U_{N,k}\|_{\ell^2} \longrightarrow 1. \tag{11}
$$

For a self-adjoint finite-dimensional matrix, a normalized vector with residual at most ρ places an eigenvalue within ρ of its proposed energy. Apply this to (11), and use the indexed convergence (4) and oscillator spectral gaps to identify the nearby eigenvalue as $\varepsilon_{N,k}$. In fact, the same spectral gap yields control of the eigenvector itself: if $v_{N,k}$ is its normalized eigenvector and $\widehat{U}_{N,k} = U_{N,k}/\|U_{N,k}\|$, then there is a choice of sign such that

$$\min_{\sigma\in\{+1,-1\}} \|\widehat{U}_{N,k} - \sigma v_{N,k}\|_{\ell^2} = O_k(h^6).$$

To see this, use the convergent separation of the adjacent discrete energies to bound the spectral projection onto the orthogonal complement of $v_{N,k}$ by the residual in (11) divided by a fixed positive gap. This statement has fixed-index scope only. It follows that

$$\varepsilon_{N,k} = \varepsilon_k + h^2 p_k + h^4 q_k + O_k(h^6).$$

Since $h^2 = N^{-1}$ and $\lambda_{N,k} = N/2 - \varepsilon_{N,k}$, equation (1) follows. □

6. Relationship to the sine-kernel reference and its limits

The half-filled finite sine kernel has entries

$$(B_N)_{j\ell} = \frac{1}{2\pi}\int_{-\pi/2}^{\pi/2} e^{i(j-\ell)\theta}\, d\theta.$$

For $W = 1/4$, the reference matrix B_N is Slepian's finite concentration matrix $p(N, W)$, equation (21), and $J_N = \frac{2}{N}\sigma(N, 1/4)$ in terms of his commuting tridiagonal matrix, equation (14). Slepian proves that the ordered eigenvectors of $\sigma(N, W)$ have strictly decreasing finite concentration eigenvalues, §4.1, equations (129)–(131), pp. 1403–1404. The argument uses Sturm zero ordering and continuity in the bandwidth; **commutation alone would not imply the index matching**. Thus the k-th largest Jacobi eigenvector corresponds to the k-th largest occupation of B_N. With this classical ordering input, Slepian's fixed-index concentration asymptotic (reproduced by Adcock, Huybrechs and Martín-Vaquero, §3.2, equation (3.10)) and the scalar endpoint expansion give, for fixed k, the modular residual limit

$$\lim_{N\to\infty} e_{N,k} = \frac{1}{2}\log\pi - \log(k!) - k\log 2 + \left(k + \tfrac{1}{2}\right)[\log(2k+1) - 1].$$

The extra inverse-square Jacobi coefficient in (1) **does not** imply a next-order modular residual: a compatible quantitative correction to the discrete-prolate concentration asymptotic would also be required. Nothing in this manuscript establishes a bound uniform for $k = k(N)$, the operator-norm bound $\|E_N\| = O(1)$, or the required Wilson longitudinal modular trace estimate $\mathrm{Tr}(S_N E_N) = o_\mu(\log N)$. The separate whole-line argument in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* establishes the fixed-positive-mass leading entropy law with bounded remainder. The fixed-index analysis here does not supply the whole-line argument. Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy* determines the finite part for fixed $\mu > 0$, and Theorem 3 of *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals* proves $\|E_N\| \le 7$ for even $N \ge 2$. Neither statement turns this fixed-index expansion into a moving-index expansion.

References

- D. Slepian, "Prolate spheroidal wave functions, Fourier analysis, and uncertainty V: The discrete case," *Bell System Technical Journal* **57** (1978), 1371–1430, DOI: 10.1002/j.1538-7305.1978.tb02104.x; see equation (14), equation (21), and §4.1, equations (129)–(131), pp. 1403–1404.

- F. A. Grünbaum, “Eigenvectors of a Toeplitz matrix: discrete version of the prolate spheroidal wave functions,” *SIAM Journal on Algebraic Discrete Methods* **2** (1981), 136–141, DOI: 10.1137/0602017.
- B. Adcock, D. Huybrechs, and J. Martín-Vaquero, “On the numerical stability of Fourier extensions,” *Foundations of Computational Mathematics* **14** (2014), 635–687; preprint ht tps://arxiv.org/html/1206.4111v3, §3.2, equation (3.10).

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